

Abstract

Agentic Digital Twins (ADTs) represent the vanguard of industrial asset management, automating diagnostic, prognostic, and maintenance scheduling tasks via Tool-augmented Large Language Models (LLMs). However, deploying ADTs in safety-critical industrial settings is hindered by the lack of reliable uncertainty quantification. Standard LLMs are notoriously overconfident due to RLHF alignment, leading to silent, catastrophic failures. In this paper, accepted/submitted for IEEE Smart World Congress (SWC 2026) / Digital Twin 2026, we present a systematic trajectory calibration study on the PHMForge benchmark. We formulate a 6-dimensional diagnostic feature vector $f(\tau)$ capturing early token generation entropy, confidence gradients, Model Context Protocol (MCP) tool execution telemetry, and action loop ratios. We implement and compare five calibration schemes across open-weights and commercial models. Our empirical feature importance analysis reveals that logprob variance (30.0%) and MCP error ratio (28.7%) dominate failure prediction, whereas verbalized model confidence accounts for under 4%. Furthermore, we demonstrate that selective abstention reduces Expected Operational Costs (EOC) by up to 83%.

1 Introduction

Industrial Digital Twins have transitioned from static physical representations to Stage 4 Autonomous Management systems, powered by LLM-based agentic workflows. In these environments, agents receive high-level requests (e.g., prognosticating bearing lifetimes or optimizing maintenance windows) and interact with safety-critical machinery

through the Model Context Protocol (MCP). Under this paradigm, agents select, configure, and run diagnostic toolchains, interpreting multi-sensor outputs dynamically.

Despite their versatility, current SOTA agents are prone to hallucinating tool parameters and producing overconfident, incorrect predictions. In domain-critical assets, such as aerospace propulsion or power grid turbine maintenance, un-

monitored failure is unacceptable. Hence, equipping agentic twins with the ability to quantify their uncertainty and selectively abstain from actions when confidence is low is a crucial step towards safe industrial deployment.

2 Methodology and Trajectory Diagnostics

Let τ be the execution trajectory of an agent solving an industrial maintenance scenario S . The trajectory is defined by a sequence of thought, action, and observation steps: $\tau = (s_0, a_0, o_0, s_1, a_1, o_1, \dots, s_T)$.

2.1 Trajectory Diagnostic Feature Vector $f(\tau)$

We extract a 6-dimensional diagnostic feature vector $f(\tau) \in \mathbb{R}^6$:

$$f(\tau) = [\mathcal{H}_{\text{early}}, \nabla \text{Conf}, \text{Ratio}_{\text{err}}, \text{Var}(\text{LogProbs}), \mathcal{S}_{\text{verbalized}}, \text{Ratio}_{\text{loop}}] \quad (1)$$

where $\mathcal{H}_{\text{early}}$ measures initial step entropy, $\text{Ratio}_{\text{err}}$ is the fraction of failed MCP tool calls, $\text{Var}(\text{LogProbs})$ measures token uncertainty along the trajectory, and $\text{Ratio}_{\text{loop}}$ captures repetitive action looping behaviors.

2.2 Expected Operational Cost (EOC)

To measure the economic impact of calibration in

industrial plants, we define the Expected Operational Cost (EOC):

$$EOC = C_{\text{inspection}} \cdot N_{\text{false_alarm}} + C_{\text{catastrophic}} \cdot N_{\text{unnoticed_failure}} + C_{\text{review}} \cdot N_{\text{abstain}} \quad (2)$$

where $C_{\text{inspection}} = \$1,000$, $C_{\text{catastrophic}} = \$50,000$, and $C_{\text{review}} = \$500$.

3 Experiments and Results

We evaluate five calibration models across 7 LLM backbones (including open-weights models like Llama-4 Maverick and commercial models like GPT-4o-mini).

3.1 Calibration Performance Comparison

Table 1 summarizes ECE ($\downarrow$) and AUROC ($\uparrow$) performance.

3.2 Feature Importance Analysis

Empirical feature importance ranking across all backbones demonstrates:

- **Logprob Variance (30.03%):** Primary predictor of internal cognitive dissonance.
- **MCP Error Ratio (28.70%):** Direct operational friction signal.
- **Early Entropy (22.71%):** Initial confusion prior to tool execution.
- **Action Loop Ratio (14.58%):** Diagnostic indicator of agent state-machine trapping.

Uncertainty Quantification and Selective Abstention for Agentic Digital Twins: A Trajectory Calibration Study on PHMForge

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- **Verbalized Confidence (3.98%):** Confirms that stated confidence is largely uninformative.

3.3 MCP Tool Dichotomy

Tool telemetry shows a sharp dichotomy: Data Retrieval Tools (`load_dataset`, `web_search`) generate high verbalized confidence (0.85–0.95) regardless of eventual task success, whereas Analytical Tools (`train_rul_model`, `calculate_mae`) impose strict validation constraints that expose true model uncertainty via MCP errors.

3.4 Economic Impact: *EOC* Reduction

Applying HTC Selective Abstention at threshold $\theta \geq 0.75$ reduces the Expected Operational Cost (*EOC*) from \$12,400 (under raw execution) to \$2,100 per 100 runs—achieving an **83% operational cost reduction**.

4 Conclusion

Our trajectory calibration framework for Agentic Digital Twins establishes that internal cognitive telemetry and MCP execution signals drastically outperform verbalized confidence. This provides a robust foundation for safe, cost-optimized deployment in industrial environments.